Conveyor Line

Scenario A, Round 1 - 60 minutes, teams of 4 to 5

Your task

Your team builds three stations of a small conveyor line. The detection station notices when a part arrives. The sorting station checks whether a part is bright or dark and sends it to lane A or lane B. The counting station counts the finished parts.

In this round every station works on its own. No station needs a signal from another station. Build each station, test it on its own and show all three working at the end.

What the system must do

Detection station

- The ultrasonic sensor checks whether a part is closer than a set distance.
- The detection LED is on while a part is there.

Sorting station

- When the sort button is pressed, the light sensor reads the part.
- A bright part lights lane A, a dark part lights lane B.
- The result appears in the Serial Monitor.

Counting station

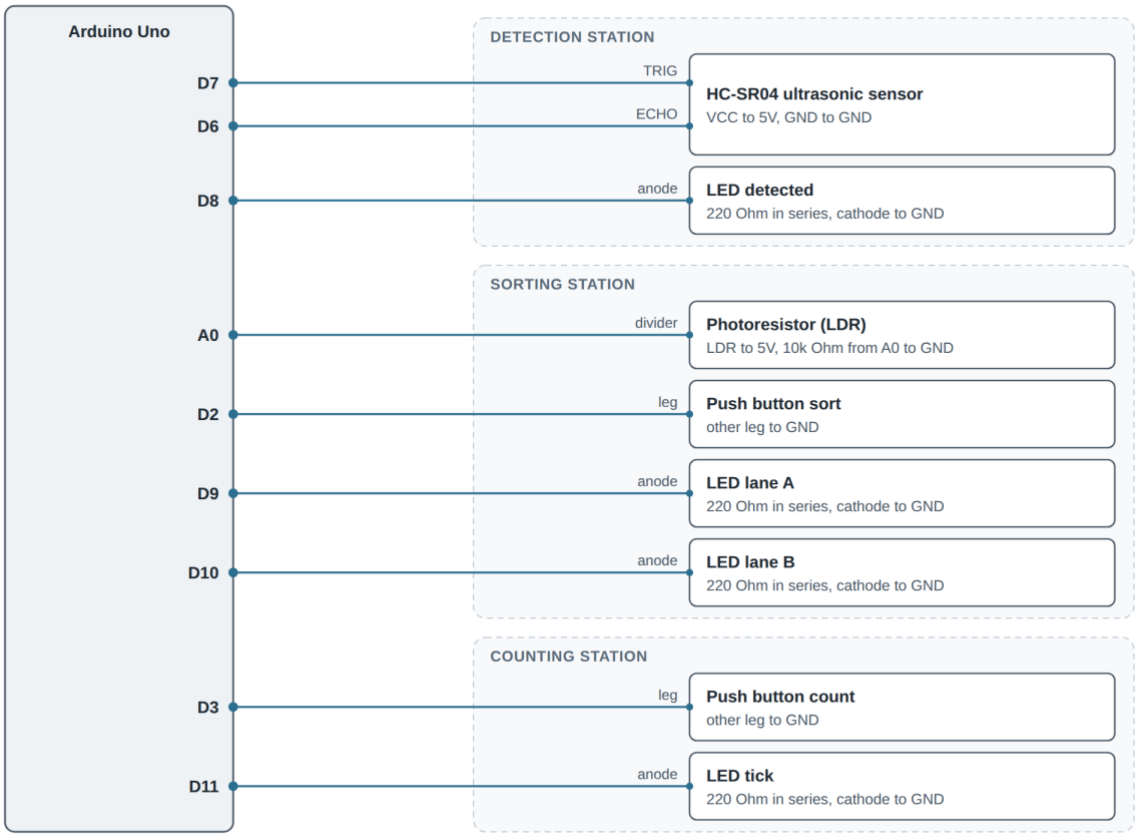
- Each press of the count button adds one part.
- The tick LED blinks once for each part.
- The current count appears in the Serial Monitor.

Parts

Part	Quantity
Arduino Uno and breadboard	1
HC-SR04 ultrasonic sensor	1
Photoresistor (LDR)	1
10k Ohm resistor	1
Push button	2
LED	4
220 Ohm resistor	4
Jumper wires	about 20

Wiring

Scenario A Conveyor line (same wiring in both rounds)



Use the pins shown in the diagram. Every LED needs a 220 Ohm resistor in series.

The code

You can build the circuit on a real board or in TinkerCad Circuits. In TinkerCad, open the code editor, switch it to Text and paste the code. Fill in every blank marked ___ before you start the program. The code you get is not fully correct. When the system does not do what the list above asks, find the cause and fix it.

Blank	What to fill in
NEAR_CM	Distance in centimeters below which a part counts as present, for example 15.
SORT_THRESHOLD	Light value between 0 and 1023 that separates bright from dark. Start with the middle and adjust it after testing.
Detection LED	The two values that switch the LED on when a part is there and off when it is gone.

```
/*
 * Scenario A  Conveyor line
 * Round 1  Three independent stations
 * Fill in every ___ before you upload.
 */
```

```
// Pin map (identical in Round 1 and Round 2)
const int PIN_TRIG         = 7;   // ultrasonic sensor, detection station
const int PIN_ECHO         = 6;
const int PIN_LED_DETECTED = 8;
const int PIN_LDR          = A0;  // light sensor, sorting station
const int PIN_LED_LANE_A   = 9;
const int PIN_LED_LANE_B   = 10;
const int PIN_BTN_SORT     = 2;
const int PIN_BTN_COUNT    = 3;   // counting station
const int PIN_LED_TICK     = 11;

// Constants
const int NEAR_CM        = ___;  // a part is present when nearer than this (cm)
const int SORT_THRESHOLD = ___;  // above means bright part, lane A (0 to 1023)

int  partCount = 0;
bool sortPrev  = true;   // last button reading, used for edge detection
bool countPrev = true;

long readCM() {
  digitalWrite(PIN_TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(PIN_TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(PIN_TRIG, LOW);
  long d = pulseIn(PIN_ECHO, HIGH, 20000);
  return (d == 0) ? 999 : d * 0.034 / 2;
}

// Detection station
void stationDetection() {
  digitalWrite(PIN_LED_DETECTED, readCM() < NEAR_CM ? ___ : ___);
}

// Sorting station
void stationSorting() {
  bool pressed = digitalRead(PIN_BTN_SORT) == LOW;
  if (pressed && sortPrev) {
    int v = analogRead(PIN_LDR);
    bool laneA = v < SORT_THRESHOLD;
    digitalWrite(PIN_LED_LANE_A, laneA ? HIGH : LOW);
    digitalWrite(PIN_LED_LANE_B, laneA ? LOW : HIGH);
    Serial.println(laneA ? "Sorted to lane A" : "Sorted to lane B");
  }
  sortPrev = !pressed;
}

// Counting station
void stationCounting() {
  bool pressed = digitalRead(PIN_BTN_COUNT) == LOW;
  if (pressed && countPrev) {
    partCount++;
    digitalWrite(PIN_LED_TICK, HIGH); delay(80); digitalWrite(PIN_LED_TICK, LOW);
    Serial.print("Count "); Serial.println(partCount);
  }
  countPrev = !pressed;
}

void setup() {
  pinMode(PIN_TRIG, OUTPUT); pinMode(PIN_ECHO, INPUT);
  pinMode(PIN_LED_DETECTED, OUTPUT);
  pinMode(PIN_LED_LANE_A, OUTPUT); pinMode(PIN_LED_LANE_B, OUTPUT);
  pinMode(PIN_BTN_SORT, INPUT_PULLUP);
  pinMode(PIN_BTN_COUNT, INPUT_PULLUP);
  pinMode(PIN_LED_TICK, OUTPUT);
  Serial.begin(9600);
  Serial.println("Conveyor stations ready (Round 1)");
}

void loop() {
  stationDetection();
  stationSorting();
  stationCounting();
```

```
  delay(50);
}
```

How you work

Every member of the team takes one role.

Role	Responsibility	Works closely with
Line coordinator	Splits the work, keeps track of time and signs off the result.	Everyone
Detection developer	Builds and tests the detection station.	Tester
Sorting developer	Builds and tests the sorting station.	Tester
Counting developer	Builds and tests the counting station.	Tester
Tester	Tests every station against the list above and reports what does not work.	Everyone

In a team of four, the line coordinator also takes the role of tester.

Apart from the roles there are no rules for this round. Your team decides how you talk to each other.

You have 60 minutes.

After the round

Take about 10 minutes as a team and talk about these questions.

- How much did the three stations depend on each other?
- Could each station be built and tested without waiting for someone else?
- How did you decide who works on what?
- When something did not work, how did you find the cause?
- Where did you need to talk to each other, and where not?
- What would you change if you did this again?